

ECON 498—ADVANCED SPECIAL TOPICS IN ECONOMICS (GENERAL EQUILIBRIUM)

Spring 2026

Department of Economics
Nazarbayev University

Instructor: Alejandro Melo Ponce, 8.510, alejandro.melo@nu.edu.kz

Lecture time/Location: TBA (two weekly sessions of 75 minutes, or one long weekly session, 150–180 minutes).

Office hours: By appointment; details to be announced.

Course Description

This course provides a mathematically rigorous, PhD-level treatment of general equilibrium theory. We develop the Walrasian model in exchange and production economies, study welfare and efficiency under standard assumptions, and establish equilibrium existence using fixed-point methods and core tools from convex and variational analysis.

A central component of the course is the cooperative foundations of competitive equilibrium. We analyze the core of an economy and its convergence to competitive allocations in large economies, and we study equilibrium in environments with nonconvexities and indivisible goods. These settings naturally connect general equilibrium analysis to assignment problems and auction-theoretic mechanisms.

If time permits, selected advanced topics include equilibrium under uncertainty and economies with a continuum of agents. Throughout, the course emphasizes formal reasoning, careful modeling, and conceptual clarity, and is intended for students with strong interests in economic theory, a good mathematical background, and an interest in developing mathematical maturity.

Course Requirements

Official prerequisites. Enrollment requires instructor approval. In addition, students must have completed *at least one* of the following courses with a grade of C– or higher:

- ECON 201—Intermediate Microeconomics
- ECON 202—Intermediate Macroeconomics
- MATH 321—Probability
- MATH 322—Mathematical Statistics

Expected preparation. This is an advanced, proof-based course. Students are expected to be comfortable with rigorous mathematical reasoning and with reading and constructing formal arguments. Familiarity with real analysis and related mathematical tools is helpful but not required; the necessary analytical concepts will be introduced and developed as part of the course. Students who would like a structured review may consult Green and Heller (1981). An optional math bootcamp lecture may be offered at the start of the term depending on enrollment.

Course Aims

1. Develop a deep understanding of the formal structure of general equilibrium models under exchange and production.
2. Equip students with the mathematical tools required to analyze existence, efficiency, and welfare properties of competitive equilibria.
3. Examine the relationship between competitive equilibria and cooperative solution concepts, particularly the core and core convergence.

4. Connect general equilibrium theory to modern applications, including indivisible goods and auction-based allocation mechanisms.
5. Foster the ability to read, interpret, and construct rigorous theoretical economic arguments, and to engage with selected advanced extensions of general equilibrium theory.

Course Learning Outcomes

By the end of the course the student will be expected to be able to

1. **Analyze** exchange and production economies using the formal framework of general equilibrium theory, including selected advanced environments such as uncertainty and large economies.
2. **Construct and reproduce** key existence proofs for Walrasian equilibrium, including proofs based on excess demand, fixed-point theorems, and quasiequilibrium methods.
3. **Evaluate** the welfare properties of competitive equilibria and formally apply the First and Second Welfare Theorems under stated assumptions.
4. **Compare and interpret** competitive equilibria and core allocations, and **explain** the logic of core convergence in large economies.
5. **Apply** general equilibrium tools to economies with indivisible goods and **relate** equilibrium concepts to auction and combinatorial allocation settings.

Learning and Teaching Methods

Lectures, guided reading of selected notes and papers, and optional problem sets for self-study (not graded).

Course Textbooks

There is no single required textbook, but the course is anchored on:

- Andreu Mas-Colell et al. (1995). *Microeconomic Theory*. New York: Oxford University Press, General Equilibrium chapters. For historical references, consult Arrow and Debreu (1954), McKenzie (1959), and Debreu (1959).
- Mathematical core: Green and Heller (1981), Rockafellar (1970), Border (1985), and Aliprantis and Border (2006)
- Instructor notes.
- Selected papers as assigned: Anderson (1978), Gul and Stacchetti (1999), and Debreu and Scarf (1963)

You will be responsible for all material covered or assigned in class. Therefore, textbooks and notes are complements to lectures and are not a substitute for them.

Course Website

Additional course material, such as lecture notes and announcements will be posted on the Moodle page of the course, which you should check regularly.

Assessments

Midterm 1 Exam covering the mathematical toolkit and exchange economies. The exam is held outside regular class time (typically Friday/weekend); the exact date is announced during the term.

Midterm 2 Exam covering welfare results and equilibrium analysis with production. The exam is held outside regular class time (typically Friday/weekend); the exact date is announced during the term.

Midterm 3 Exam covering equilibrium existence with production, the core and core convergence, and economies with indivisible goods. The exam is held during the official Finals Week; the exact date is announced by the Registrar.

Table 1: Assessment weights

Assessment component	Weight
Midterm 1 (outside class time; Friday/weekend)	25%
Midterm 2 (outside class time; Friday/weekend)	25%
Midterm 3 (held during Finals Week; outside class time)	25%
Final Project (due during Finals Week)	25%

Final Project Exploratory project on an advanced topic or paper (e.g., indivisibilities/auctions, uncertainty/Arrow–Debreu, sequential markets, continuum agents). Emphasis is on synthesis, formal reasoning, and conceptual clarity. Detailed guidelines will be provided during the term.

I will compute your final numeric grade by computing the weighted average of the aforementioned components according to the weights given in the table above.

Grading

Your course grade is determined by the weighted average of the assessments above according to the following scale:

A	95–100	A–	90–94.99
B+	85–89.99	B	80–84.99
B–	75–79.99	C+	70–74.99
C	65–69.99	C–	60–64.99
D+	55–59.99	D	50–54.99
F	0–49.99		

Course Expectations

- Class will begin promptly at the scheduled starting time. As a courtesy to others, it is requested that you not arrive after class has begun or leave before it ends. PLEASE NOTE: Since showing up late and leaving early is disruptive to the class, students who disrupt the class in this or any other manner may be administratively dropped or have their course grade reduced at the instructor's discretion. If you must leave class early or will be late, please let me know before class begins.
- Exams* will be held outside of class, typically on Fridays or weekends. The dates of the midterm exams are fixed, but their locations will be announced at a later date. The final exam will be scheduled by the Office of the Registrar.

If you miss an exam, you will receive a grade of zero unless you provide a valid and documented reason. In the case of a missed midterm exam with valid documentation, the weight of that exam will be uniformly allocated to the other midterms. Appeals for midterm exam 1 and 2 must be submitted in writing within five days of receiving the graded exam. Appeals for midterm exam 3 or the final course grade must also be submitted in writing within two days of the grades being posted.

- The Nazarbayev University Special Learning Needs Committee (SLNC) is committed to creating an equitable and inclusive education environment for all students. If you have a qualified special learning need (physical, cognitive, socio-emotional, and psychological), please contact the SLNC as early as possible to ensure you receive the fullest support available. If you already have approved SLNC accommodations, please share them with your teaching faculty as soon as possible. Accommodations cannot be applied retroactively, and will only be active once your teaching faculty has received them. ¹

Academic Misconduct

Students are expected to act with integrity. Cheating and plagiarism will not be tolerated. Any student deemed academically dishonest will be reported to the SSH disciplinary committee. Any cases of academic misconduct will

¹This includes conditions that may be physical, cognitive, socio-emotional, and psychological. For more information, please contact us at SLNC@nu.edu.kz and follow this link to the Accommodation Request Instructions.

be handled in accordance with the procedures of the Student Code of Conduct. The penalties may include anything from failing grade on a particular assignment to a loss of stipend, and even, dismissal from the University. You can read more about academic misconduct in the Student Code of Conduct. https://aao.nu.edu.kz/academic_integrity

AI Policy

This policy applies to any generative AI tool and all AI-generated products. You are discouraged from using AI tools unless explicitly instructed by me. If you are unsure or have questions, you must contact me before using AI for any assignment.

In this course, the potential risks of AI usage generally outweigh the benefits to student learning. If AI use is permitted, you are required to clearly indicate which parts of your assignment were created using AI and which parts are your original work. Additionally, no more than 25% of an assignment may be generated with AI, even when its use is approved.

You are required to sign an AI usage contract confirming your understanding and agreement to these policies. Unacknowledged use of AI tools will be treated as plagiarism, and penalties will be applied in accordance with the NU Student Code of Conduct.

Tentative Course Outline

Session	Topics and Assignments	Course Aims	CLOs
Week 1	Math bootcamp: metric spaces and some notions of topology.	1,2,5	1,2
Week 2	Convexity and aggregation: <i>Carathéodory</i> and <i>Shapley–Folkman</i> .	2,3	2,4
Week 3	Correspondences and fixed points: hemicontinuity; Berge’s theorem; Brouwer/Kakutani/KKM.	2	2
Week 4	Exchange economies I: preferences, budgets, demand, Walras’ law, excess demand.	1,2	1,2
Week 5	Exchange economies II: Walrasian equilibrium; welfare and supporting prices; welfare theorems.	1,2	1,3
Week 6	Existence in exchange economies (fixed-point route); Midterm 1 (Friday/weekend, outside class time).	1,2	1,2,3
Week 7	The core: blocking, core allocations, core vs competitive equilibrium; Edgeworth conjecture.	3	4
Week 8	Spring Break		
Week 9	Production economies: firms, technologies, profit maximization; equilibrium with production; welfare.	1,2	1,3
Week 10	Existence with production I: quasiequilibrium; free disposal; truncation arguments.	1,2	2
Week 11	Existence with production II: generalized games and fixed-point construction. Midterm 2 (Friday/weekend, outside class time)	1,2	2
Week 12	Core convergence Anderson (1978) and approximate convexification.	3,5	4
Week 13	Indivisibilities and equilibrium Gul and Stacchetti (1999); links to auctions and assignments.	4,5	5
Week 14	Selected topics: Arrow–Debreu uncertainty; sequential markets; continuum agents. Project workshop.	5	1
Finals Week	Midterm 3 and Final Project due.	1–5	1–5

Disclaimer

Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.

References

Aliprantis, Charalambos D. and Kim C. Border (2006). *Infinite Dimensional Analysis: A Hitchhiker’s Guide*. Springer.
 Anderson, Robert M. (1978). “An Elementary Core Equivalence Theorem”. In: *Econometrica* 46.6, pp. 1483–1487.
 ISSN: 00129682, 14680262. URL: <http://www.jstor.org/stable/1913840> (visited on 12/16/2025).

- Arrow, Kenneth J. and Gerard Debreu (1954). "Existence of an Equilibrium for a Competitive Economy". In: *Econometrica* 22.3, pp. 265–290. ISSN: 00129682, 14680262. URL: <http://www.jstor.org/stable/1907353> (visited on 12/16/2025).
- Border, Kim C. (1985). *Fixed point theorems with applications to economics and game theory*. Cambridge university press.
- Debreu, Gerard (1959). *Theory of value: An axiomatic analysis of economic equilibrium*. Vol. 17. Yale University Press.
- Debreu, Gerard and Herbert Scarf (1963). "A Limit Theorem on the Core of an Economy". In: *International Economic Review* 4.3, pp. 235–246. ISSN: 00206598, 14682354. URL: <http://www.jstor.org/stable/2525306> (visited on 12/16/2025).
- Green, Jerry and Walter P. Heller (1981). "Mathematical analysis and convexity with applications to economics". In: *Handbook of mathematical economics* 1, pp. 15–52.
- Gul, Faruk and Ennio Stacchetti (1999). "Walrasian equilibrium with gross substitutes". In: *Journal of Economic theory* 87.1, pp. 95–124.
- Mas-Colell, Andreu, Michael D. Whinston, and Jerry R. Green (1995). *Microeconomic Theory*. New York: Oxford University Press.
- McKenzie, Lionel W. (1959). "On the Existence of General Equilibrium for a Competitive Market". In: *Econometrica* 27.1, pp. 54–71. ISSN: 00129682, 14680262. URL: <http://www.jstor.org/stable/1907777> (visited on 12/16/2025).
- Rockafellar, R. Tyrrell (1970). *Convex Analysis*. Princeton, NJ: Princeton University Press.